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Exercise 02 - SQL Foundations

01 The Students - Table

01. SELECT DISTINCT department,
FROM students_table ;

department
IT
HR
Finance

02. SELECT department,
AVG AS avg_age
FROM students_table
GROUP BY department ;

department	avg_age
IT	20
HR	22
Finance	23

03. SELECT department,
COUNT(*) AS student_count
FROM student_table
GROUP BY department
HAVING COUNT(*) > 1 ;

department	student_count
IT	2
HR	2

04. SELECT department,
student_id,
name,
age
FROM students_table
WHERE age between 21 AND 23 ;

student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

05 SELECT student_id,
name,
age,
department
FROM students_table
WHERE department IN HR, IT AND
age > 21 ;

student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

02 The Courses Table

06. SELECT department,
SUM(credits) AS total_credits
FROM courses_table
GROUP BY department
HAVING SUM(credits) > 5;

department	credits
IT	11

07 SELECT course_id,
course_name,
department,
credits
FROM course_table
WHERE credits < 4;

course_id	course_name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

08. SELECT course_id,
course_name,
credits
FROM course_table
ORDER BY credits DESC
LIMIT 3

course_id	course_name	department	credits
102	Python	IT	4
103	Data Science	IT	4
101	SQL Basics	IT	3

~~08~~ 03.

03. The Enrollments Table

09. SELECT MAX(grade) AS max_grade,
min(grade) AS min_grade,
AVG(grade) AS avg_grade,
FROM enrollments_table;

max_grade	min_grade	avg_grade
90	78	84.6

10. SELECT course_id
COUNT(*) AS total_enrollments
FROM enrollments_table
GROUP BY course_id;

course_id	total enrollments
101	1
102	1
103	1
104	1
105	1

Q4 The Salaries Table

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11. SELECT department,
      SUM(salary) AS total_salary,
      SUM(bonus) AS total_bonus
FROM   employees salaries_table
GROUP BY department

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department	total_salary	total_bonus
IT	122 000	10 500
HR	109 000	7 500
Finance	10 000	6 000

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12 SELECT employee_id,
      name,
      salary,
      bonus,
      (Salary + bonus) AS total_compensation
FROM   salaries_table
WHERE  (salary + bonus) > 6000 ;

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employee_id	name	salary	bonus
1	Tom	60 000	5000
3	Tyke Spike	70 000	6000
4	Batch Tyke	62 000	5500

as The Projects Table

14. SELECT department, project_id
 SUM(budget) AS total_budget
 AVG(budget) AS avg_budget
 FROM projects_table
 GROUP BY department
 HAVING AVG(budget) > 70 000;

department project_id
 14

department	total_budget	avg_budget
IT	270 000	135 000
Finance	80 000	80 000

15 SELECT project_id,
 project name,
 department,
 budget
 FROM projects_table
 WHERE budget between 50 000 AND 120 000
 AND department <> 'Marketing';

project_id	project_name	department	budget
1	AI App	IT	120 000
2	Payroll System	Finance	80 000
5	HR Portal	HR	50 000.